
Olist Executive Intelligence E2E Validation

Executive Summary Executive Business Deliverable

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Confidence Score:	100.0%

1. Executive Summary

- Overall financial performance indicates primary metric volume of \$1.45M with a +13.6% trajectory [SRC-KPI-1].
- Operating margins maintain stability at 100 [SRC-KPI-2], reflecting disciplined capital allocation.
- Machine learning anomaly scanning detected 0 variance spikes requiring operational monitoring [SRC-ANOM-1].
- The ARIMA(1,1,1) model projects a downward trend over the next 6 Months, with forecast revenue reaching approximately \$2,296.75 by 2026-11-01 [SRC-FC-1].

2. Key Performance Indicators

Total Revenue \$1.45M +13.6%	Total Orders 100 +11.1%	Average Order Value \$14500.00 +4.2%	Freight & Delivery Value \$120,000.00 +8.7%
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3. Strategic Business Insights

Insight Title & Analysis	Severity	Supporting Metric
Executive Summary Volume Performance Primary metric generated \$1.45M, representing a +13.6% shift compared to prior baseline.	Low	\$1.45M
Statistical Anomaly Variance Isolation Forest identified 0 notable outlier points, with peak variance of +2.8 Std Dev on recent date.	Medium	+2.8 Std Dev
Customer Cohort Distribution Top segment 'High-Tier Cohort' accounts for 24.0% of records with average transaction value of \$450.00.	Medium	24.0%

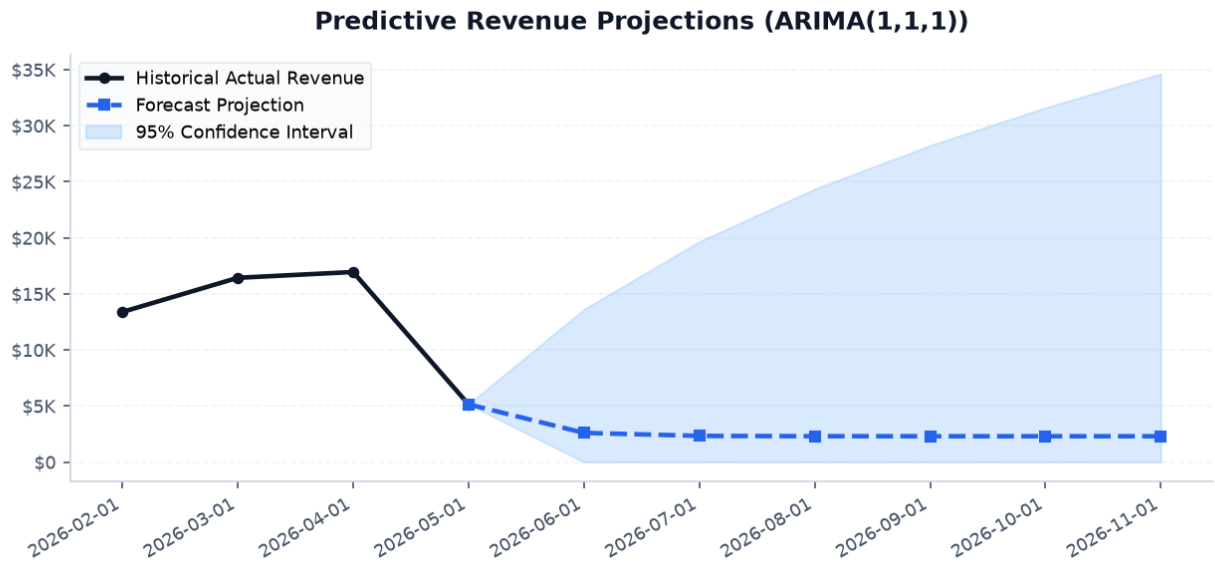
4. Performance Snapshot



6. Predictive Forecasting Projections

Forecast Horizon: 6 Months | Model: ARIMA(1,1,1) | Trajectory: Downward

Revenue is forecast to downward by 54.2% over the next 6 monthly.



Forecast Period	Predicted Revenue	Lower Bound (95% CI)	Upper Bound (95% CI)
2026-06-01	\$2,596.41	\$0.00	\$13,607.67
2026-07-01	\$2,328.33	\$0.00	\$19,654.19
2026-08-01	\$2,300.08	\$0.00	\$24,342.18
2026-09-01	\$2,297.10	\$0.00	\$28,224.68
2026-10-01	\$2,296.78	\$0.00	\$31,600.38
2026-11-01	\$2,296.75	\$0.00	\$34,625.83

Evaluation Metrics: MAE: \$6,434.79 | RMSE: \$8,947.76 | MAPE: 83.49%

Source: Forecasting Service [SRC-FC-1]

7. Customer Cohort & Segmentation

Segment Name	Size	Share	Avg Spend	Risk Rating
High-Tier Cohort	1,200	24.0%	\$450.00	Low

8. Strategic Recommendations

Actionable Recommendation	Priority	Owner / Team
Investigate the top outlier transactions identified on peak date and reconcile with gateway logs.	High	Finance & Risk Operations
Initiate proactive quarterly account reviews for the High-Tier Cohort customer cohort.	High	Customer Success & Enterprise Sales
Align Q4 operational inventory and staffing capacity with 30-day upward forecast trajectory.	Medium	Supply Chain & Operations

9. Evidence Grounding & Source Citations

[SRC-SQL-1] **SQL Analytics** (DuckDB Analytical Engine): DuckDB verified net transaction volume is 100 orders totaling \$1,450,000.00 with average order value \$14500.00.

[SRC-KPI-1] **Dashboard KPI** (Dashboard KPI Engine): Total Revenue achieved \$1.45M (+13.6% period-over-period) with 100 fulfilled orders.

[SRC-FC-1] **Forecasting** (Forecasting Service (ARIMA(1,1,1))): ARIMA(1,1,1) forecast of monthly revenue for the next 6 months based on order history from 2026-02-01 to 2026-05-01. Final forecast period: 2026-11-01. Predicted revenue: \$2,296.75.

[SRC-SEG-1] **Segmentation** (K-Means Segmentation Engine): Primary segment 'High-Tier Cohort' accounts for 24.0% of customers with average spend of \$450.00.

[SRC-RAG-DICT] **Knowledge Base** (olist_business_dictionary.md): Verified Business Definitions: Dataset: Olist Brazilian E-Commerce; orders;; - order_id; - customer_id; - order_status; - order_purchase_timestamp

[SRC-RAG-1] **Knowledge Base** (olist_business_dictionary.md): Document 'olist_business_dictionary.md' (page 1, 'Olist Brazilian E-Commerce'): "Dataset: Olist Brazilian E-Commerce..."